On the integrality of some P-recursive sequences

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Joint work with Alin Bostan

Outline

① Introduction and definitions

② Motivating example

③ Generalization

④ Algorithm

⑤ Conclusion

Key definitions

Definition

A sequence $(s_n)_{n\in\mathbb{N}}$ of rational numbers is called *P-recursive* if it satisfies a linear (homogeneous) recurrence relation with coefficients in $\mathbb{Q}[n]$.

$$P_0(n)s_n + P_1(n)s_{n-1} + \ldots + P_k(n)s_{n-k} = 0, \quad P_0 \neq 0,\ n \geq k.$$

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Examples:

- *Fibonacci numbers*: $F_n = F_{n-1} + F_{n-2}, \quad F_0 = F_1 = 1.$
- *Catalan numbers*: $(n+1)C_n = 2(2n-1)C_{n-1}, \quad C_0 = 1.$
- *Motzkin numbers*:

$$(n+2)m_n = (2n+1)m_{n-1} + (3n-3)m_{n-2}, \quad m_0 = m_1 = 1.$$

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Definition

A power series $f(x) \in \mathbb{Q}[[x]]$ is called *D-finite* if it satisfies a linear (homogeneous) differential equation with coefficients in $\mathbb{Q}[x]$.

$$a_r(x)f^{(r)}(x) + \ldots + a_1(x)f'(x) + a_0(x)f(x) = 0, \quad a_r(x) \neq 0.$$

Key definitions

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Definition

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Theorem [Stanley, 1980]

A power series is D-finite if and only if its coefficient sequence is P-recursive.

Problem setup

Let $(s_n)_n$ be a P-recursive sequence of rational numbers.

Let $S(x) = \sum_{n \geq 0} s_n x^n$ be the generating function of $(s_n)_n$.

- When are $(s_n)_n$ and $S(x)$ *integral*?
  $\rightarrow$ $s_n \in \mathbb{Z}$ for all $n \in \mathbb{N}$.
- When are $(s_n)_n$ and $S(x)$ *globally bounded*?
  $\rightarrow$ $\exists C \in \mathbb{Z}^*$ such that $C^n s_n \in \mathbb{Z}$ for all $n \geq 1$.
  $\rightarrow$ For $S(x)$: nonzero radius of convergence.
- When is $S(x)$ *algebraic*?
  $\rightarrow$ $\exists P(x, y) \in \mathbb{Q}[x, y] \setminus \{0\}$ such that $P(x, S(x)) = 0$.

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Motzkin-type sequences

A sequence $(m_n)_n$ that satisfies the recurrence relation

$$(n+2)m_n = (2n+1)m_{n-1} + (3n-3)m_{n-2}, \quad n \geq 2 \tag{1}$$

with some $m_0, m_1 \in \mathbb{Q}$ is called a *Motzkin-type sequence*.

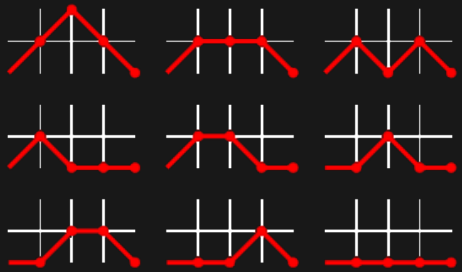
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$m_0 = m_1 = 1 \rightsquigarrow$ *Motzkin numbers*: 1, 1, 2, 4, 9, 21, 51, 127, ...



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For which (m_0, m_1) is $\sum_{n\geq 0} m_n x^n$ algebraic/globally bounded/integral?

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Theorem [Klazar, Luca, 2005]

A Motzkin-type sequence is integral if and only if $m_0 = m_1 \in \mathbb{Z}$. If $m_0 \neq m_1$, the sequence is not globally bounded.

Motzkin-type sequences

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with some $m_0, m_1 \in \mathbb{Q}$ is called a *Motzkin-type sequence*.

- $m_0 = m_1$
  - $m_0 = m_1 = 1 \implies$ Motzkin numbers.
  - $m_0 = m_1 \in \mathbb{Z} \implies$ Motzkin numbers, scaled.
- $m_0 \neq m_1$
  - Convert (1) to a first-order inhomogeneous ODE satisfied by $\sum_{n\geq 0} m_n x^n$.
  - Study one cleverly chosen pair (m_0, m_1).
  - Show that global boundedness fails.

Diagonals of multivariate rational functions (Christol, Deligne, Lipschitz...)

Definition

Let $R = \sum_{n_1,\dots,n_k \geq 0} c(n_1,\dots,n_k) x_1^{n_1} \dots x_k^{n_k} \in \mathbb{Q}[[x_1,\dots,x_k]] \cap \mathbb{Q}(x_1,\dots,x_k)$.

The *diagonal* of R is a univariate power series defined by

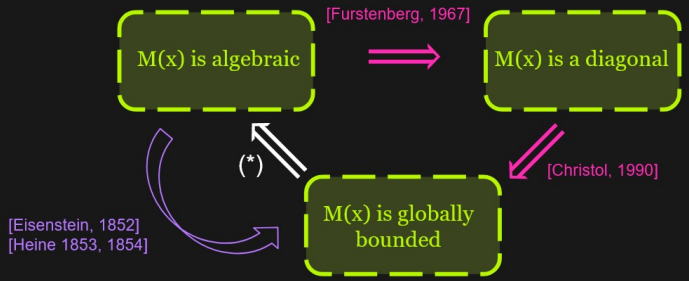
$$\mathrm{Diag}(R) = \sum_{n \geq 0} c(n,\dots,n) t^n \in \mathbb{Q}[[t]].$$

Example:

$$R = \frac{1}{1 - x_1 - x_2} = \sum_{i,j \geq 0} \binom{i+j}{i} x_1^i x_2^j,$$

$$\mathrm{Diag}(R) = \sum_{n \geq 0} \binom{2n}{n} t^n = \frac{1}{\sqrt{1-4t}}.$$

Klazar-Luca theorem for Motzkin-type sequences: refinement



Pink and purple implications hold for any functions! But (*) is special.

(*) $M(x)$ is **globally bounded** $\implies$ $m_0 = m_1$. The generating function $M_{1,1}(x)$ of Motzkin numbers is **algebraic**:

$$M_{1,1}(x) = \frac{1 - x - \sqrt{1 - 2x - 3x^2}}{2x^2}.$$

Therefore, any $M_{\mu,\mu}(x) = \mu M_{1,1}(x)$ is algebraic by linearity.

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Generalization

Conjecture

Let $(s_n)_n$ be a sequence of rational numbers that satisfies a linear homogeneous recurrence relation with polynomial coefficients of degree 1:

$$\sum_{k=0}^{d}(a_k n + b_k)s_{n-k} = 0, \quad a_0 = 1,\ a_i \in \mathbb{Q} \setminus \{0\} \text{ for } i = 1, \ldots, d \quad d \in \mathbb{N}.$$

Let $S(x) = \sum_{n\geq 0} s_n x^n$ be its generating function. The following are equivalent:

1. $S(x)$ is algebraic.
2. $S(x)$ is a diagonal.
3. $S(x)$ is globally bounded.

Proved cases of the conjecture

Recurrences of **order 1** (with linear polynomial coefficients):

$$(n + b_0)s_n + (a_1 n + b_1)s_{n-1} = 0, \quad a_1 \neq 0.$$

Closed-form expression of the generating function $S(x) = \sum_{n \geq 0} s_n x^n$:

$$S(x) = x^{-b_0}(a_1 x + 1)^{-1+b_0-\frac{b_1}{a_1}} \left(s_0 x^{b_0} {}_2F_1 \left[\begin{matrix} b_0 & b_0 - \frac{b_1}{a_1} \\ & 1 + b_0 \end{matrix} ; -a_1 x \right] + \text{const} \right). \quad (2)$$

One can show that the hypergeometric function ${}_2F_1$ in (2) is:

- either a polynomial
- or of "height 1" (equal number of integer top and bottom parameters).

Proved cases of the conjecture

$$S(x) = x^{-b_0}(a_1x+1)^{-1+b_0-\frac{b_1}{a_1}}\left(s_0x^{b_0}{}_2F_1\left[\begin{matrix} b_0 & b_0-\frac{b_1}{a_1} \\ 1+b_0 & \end{matrix}; -a_1x\right] + \text{const}\right). \quad (2)$$

One can show that the hypergeometric function ${}_2F_1$ in (2) is:

- either a polynomial
- or of "height 1" (equal number of integer top and bottom parameters).

Lemma [Christol, 1986]

A hypergeometric function of height 1 is globally bounded iff it is algebraic.

Corollary

$S(x)$ is globally bounded iff it is algebraic, iff it is a diagonal.

Proved cases of the conjecture

Theorem [Bostan, M., 2025]

Let $(s_n)_n$ be a sequence of rational numbers that satisfies:

$$(n + b_0)s_n + (a_1 n + b_1)s_{n-1} + (a_2 n + b_2)s_{n-2} = 0, \quad a_1, a_2 \neq 0.$$

Let $S(x) = \sum_{n \geq 0} s_n x^n$ be its generating function. The following are equivalent:

1. $S(x)$ is algebraic,
2. $S(x)$ is a diagonal,
3. $S(x)$ is globally bounded,

provided that $b_2 = (2a_2 b_1 - a_1 a_2 b_0)/a_1$.

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Effective version

In practice, how to decide for which values of s_0, s_1 the generating function $S(x)$ of

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is algebraic/globally bounded?

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is algebraic/globally bounded?

Possible cases:

`Alg`: $S(x)$ is algebraic for all $(s_0, s_1) \in \mathbb{Q}^2$.

`Transc`: $S(x)$ is transcendental for all $(s_0, s_1) \in \mathbb{Q}^2 \setminus \{(0, 0)\}$.

`Mixed`: The set of pairs (s_0, s_1) such that $S(x)$ is algebraic forms a one-dimensional subspace of $\mathbb{Q}^2$.

Effective version

In practice, how to decide for which values of s_0, s_1 the generating function $S(x)$ of

$$(n + b_0)s_n + (a_1 n + b_1)s_{n-1} + (a_2 n + b_2)s_{n-2} = 0$$

is algebraic/globally bounded?

$\rightsquigarrow$ **Algorithm:**

Input: b_0, a_1, b_1, a_2, b_2 such that $a_1, a_2 \neq 0,\ b_2 = (2a_2 b_1 - a_1 a_2 b_0)/a_1,\ b_0 \in \mathbb{N}$.

Output: `Alg`; `Transc`; or `Mixed`, (s_0, s_1).

The condition $b_2 = (2a_2 b_1 - a_1 a_2 b_0)/a_1$ guarantees algebraicity $\iff$ global boundedness. So we found **all** algebraic and **all** globally bounded solutions!

Key idea: linear combination of certain integrals

The generating function $S(x) = \sum_{n \geq 0} s_n x^n$ writes as:

$$S(x) = \underbrace{x^{-b_0}(1 + a_1 x + a_2 x^2)^{\frac{(b_0-1)a_1 - b_1}{a_1}}}_{\text{algebraic}} (c_1 + b_0 s_0 I_1 + (a_1 s_0 + b_0 s_1 + b_1 s_0 + s_1) I_2)$$

with

$$I_1 = \int x^{b_0 - 1}(1 + a_1 x + a_2 x^2)^{-b_0 + \frac{b_1}{a_1}} dx,$$

$$I_2 = \int x^{b_0}(1 + a_1 x + a_2 x^2)^{-b_0 + \frac{b_1}{a_1}} dx.$$

General shape of I_1, I_2 (for $b_0 \neq 0$):

$$\int x^n (1 + a_1 x + a_2 x^2)^q dx, \quad a_1, a_2 \in \mathbb{Q}^*, \ n \in \mathbb{N}, \ q \in \mathbb{Q}.$$

Algorithm outline

1. Handle $a_1^2 - 4a_2 = 0$ or $b_0 = 0$ separately.
2. Decide algebraicity of l_1 and l_2.
3. Proceed according to the table below:

l_1	l_2	**Conclusion**
Algebraic	Algebraic	`Alg`
Algebraic	Transcendental	`Mixed`, $(b_0 + 1, -a_1 - b_1)$
Transcendental	Algebraic	`Mixed`, $(0, 1)$
Transcendental	Transcendental	Check if $b_0 s_0 l_1 + (a_1 s_0 + b_0 s_1 + b_1 s_0 + s_1) l_2$ is algebraic for some $(\tilde{s_0}, \tilde{s_1}) \in \mathbb{Q}^2 \setminus \{(0,0)\}$. Yes $\implies$ `Mixed`, $(\tilde{s_0}, \tilde{s_1})$ No $\implies$ `Transc`

Algorithm outline

$$S(x) = \mathcal{A}(x)(c_1 + b_0 s_0 l_1 + (a_1 s_0 + b_0 s_1 + b_1 s_0 + s_1) l_2), \quad \mathcal{A}(x) \text{ algebraic}$$

l_1	l_2	**Conclusion**
Algebraic	Algebraic	`Alg`
Algebraic	Transcendental	`Mixed`, $(b_0 + 1, -a_1 - b_1)$
Transcendental	Algebraic	`Mixed`, $(0, 1)$
Transcendental	Transcendental	Check if $b_0 s_0 l_1 + (a_1 s_0 + b_0 s_1 + b_1 s_0 + s_1) l_2$ is algebraic for some $(\tilde{s_0}, \tilde{s_1}) \in \mathbb{Q}^2 \setminus \{(0,0)\}$. Yes $\implies$ `Mixed`, $(\tilde{s_0}, \tilde{s_1})$ No $\implies$ `Transc`

Demonstration: a déjà vu moment

The recurrence for *Motzkin-type sequences*:

$$(n+2)m_n - (2n+1)m_{n-1} - (3n-3)m_{n-2} = 0$$

corresponds to

$$b_0 = 2,\ a_1 = -2,\ b_1 = -1,\ a_2 = -3,\ b_2 = 3.$$

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$$I_1 = \underbrace{\int x(1-2x-3x^2)^{-\frac{3}{2}}\,dx}_{\text{algebraic}}, \qquad I_2 = \underbrace{\int x^2(1-2x-3x^2)^{-\frac{3}{2}}\,dx}_{\text{transcendental}}.$$

The algorithm returns `Mixed` with $(b_0+1, -(a_1+b_1)) = (3,3)$.

Another example

The recurrence for *large Schröder numbers*:

$$(n+1)s_n - (6n-3)s_{n-1} + (n-2)s_{n-2} = 0$$

corresponds to

$$b_0 = 1,\ a_1 = -6,\ b_1 = 3,\ a_2 = 1,\ b_2 = -2.$$

$$I_1 = \underbrace{\int (1-6x+x^2)^{-\frac{3}{2}}\,dx}_{\text{algebraic}}, \qquad I_2 = \underbrace{\int x(1-6x+x^2)^{-\frac{3}{2}}\,dx}_{\text{algebraic}}.$$

The algorithm returns `Alg`, proving that $S(x) = \sum_{n\geq 0} s_n x^n$ is **always algebraic**.

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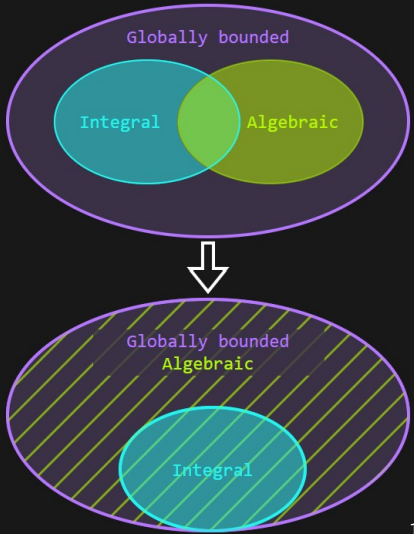
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Back to the integrality problem. Are we in $\mathbb{Z}[[x]]$?

Motivation: reduce the search for integral solutions to studying algebraic ones.

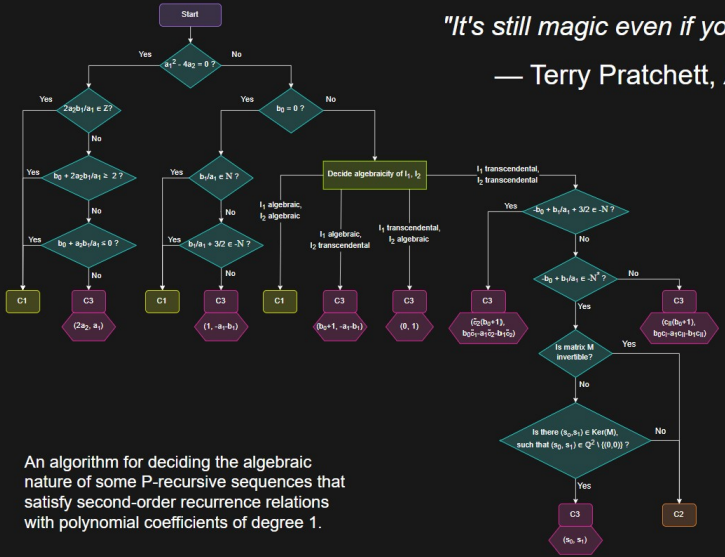
- [Fatou, 1906]:
  $\rightsquigarrow$ Integrality criteria for rational functions.
- [Pomerat, Straub, 2024]:
  $\rightsquigarrow$ Integrality criteria for n^{th} roots of power series of type $1 + x\mathbb{Z}[[x]]$.
- Work in progress by Bostan and M.:
  $\rightsquigarrow$ Integrality criteria for algebraic power series.



The road ahead

- Case $b_2 \neq \frac{2a_2b_1 - a_1a_2b_0}{a_1} \implies$ exponentials raised to non-zero powers in the generating function.
- Case $b_0 \notin \mathbb{N}$ is not currently treated by the algorithm.
- Higher-order recurrence relations.
- Nonlinear polynomial coefficients.
- Deciding when an algebraic function belongs to $\mathbb{Z}[[x]]$.

Thank you for your attention!



"It's still magic even if you know how it's done."

— Terry Pratchett, *A Hat Full of Sky*.

An algorithm for deciding the algebraic nature of some P-recursive sequences that satisfy second-order recurrence relations with polynomial coefficients of degree 1.

Outline

⑥ Appendix

Deciding the algebraic nature of I_1, I_2.

$$I_1 = \int x^{b_0-1}(1 + a_1x + a_2x^2)^{-b_0+\frac{b_1}{a_1}}\,dx, \qquad I_2 = \int x^{b_0}(1 + a_1x + a_2x^2)^{-b_0+\frac{b_1}{a_1}}\,dx.$$

General shape of I_1, I_2 (for $b_0 \neq 0$):

$$I(n, q) := \int x^n(1 + a_1x + a_2x^2)^q dx, \quad a_1, a_2 \in \mathbb{Q}^*,\ n \in \mathbb{N},\ q \in \mathbb{Q}.$$

$I(n, q)$ decomposes as a combination of:

- Algebraic functions and $I(0, q)$, or
- Algebraic functions, $I(0, q)$ and $I(-2q - 1, q)$, or
- Algebraic functions and logs multiplied by (computable) constants that depend on the recurrence parameters.

Hypergeometric functions

The *hypergeometric function* with parameters $a_1, \ldots, a_p \in \mathbb{C}, \quad b_1, \ldots, b_q \in \mathbb{C} \setminus -\mathbb{N}$ for some $p, q \in \mathbb{N}$ is defined as:

$$ {}_pF_q\left[\begin{array}{c} a_1 \ldots a_p \\ b_1 \ldots b_q \end{array}; x\right] = \sum_{n=0}^{\infty} \frac{(a_1)_n \ldots (a_p)_n}{(b_1)_n \ldots (b_q)_n} \frac{x^n}{n!}, $$

where $(k)_n$ is the rising factorial (also known as *Pochhammer symbol*):

$$ (k)_n = \begin{cases} 1 & \text{if } n = 0, \\ k(k+1)\ldots(k+n-1) & \text{if } n > 0. \end{cases} $$

Gaussian hypergeometric functions correspond to $p = 2,\ q = 1$:

$$ {}_2F_1\left[\begin{array}{cc} a_1 & a_2 \\ & b_1 \end{array}; x\right] = \sum_{n=0}^{\infty} \frac{(a_1)_n (a_2)_n}{(b_1)_n} \frac{x^n}{n!}. $$

Linearity of the coefficients: why it matters

Consider the recurrence relation for Apéry numbers

$$n^3 A_n = (34n^3 - 51n^2 + 27n - 5)A_{n-1} - (n-1)^3 A_{n-2}. \tag{2}$$

- $A_0 = 1, A_2 = 5 \rightsquigarrow$ integer (hence, globally bounded) sequence of Apéry numbers.
- The generating function of *any* sequence that satisfies (2) is transcendental. Proofs:
  - Creative telescoping. [Zagier, 1979], [Zeilberger, 1990]
  - Minimisation. [Adamczewski, Rivoal, 2018], [Bostan, Rivoal, Salvy, 2024]
    Used in the *istranscendental* function in Maple.
  - Asymptotics incompatible with algebraicity. [Flajolet, 1987]
    Flajolet criterion.

Assumption on b_2: explanation

$$> rec := \{(n+b_0)s(n)+(a_1n+b_1)s(n-1)+(a_2n+b_2)s(n-2), s(0) = s_0, s(1) = s_1\} :$$

$$> dsolve(gfun\,\text{:-}\,rectodiffeq(rec, s(n), S(x)), S(x));$$

$$S(x) = (1 + a_1x + a_2x^2)^{-\frac{b_2}{2a_2}-1+\frac{b_0}{2}} \left(\int (a_1s_0x + b_0s_1x + b_1s_0x + b_0s_0 + s_1x)(1 + a_1x + a_2x^2)^{-\frac{a_2b_0-b_2}{2a_2}} x^{b_0-1} \right.$$

$$\left. e^{\frac{\operatorname{artanh}\left(\frac{2a_2x+a_1}{\sqrt{a_1^2-4a_2}}\right)(a_1a_2b_0+a_1b_2-2a_2b_1)}{a_2\sqrt{a_1^2-4a_2}}} dx + c_1 \right) x^{-b_0} e^{-\frac{\operatorname{artanh}\left(\frac{2a_2x+a_1}{\sqrt{a_1^2-4a_2}}\right)(a_1a_2b_0+a_1b_2-2a_2b_1)}{a_2\sqrt{a_1^2-4a_2}}}.$$

Assumption on b_2: explanation

$> rec := \{(n+b_0)s(n)+(a_1n+b_1)s(n-1)+(a_2n+b_2)s(n-2), s(0) = s_0, s(1) = s_1\} :$

$> dsolve(gfun : - rectodiffeq(rec, s(n), S(x)), S(x));$

$$S(x) = (1 + a_1x + a_2x^2)^{-\frac{b_2}{2a_2}-1+\frac{b_0}{2}} \left(\int (a_1s_0x + b_0s_1x + b_1s_0x + b_0s_0 + s_1x)(1 + a_1x + a_2x^2)^{-\frac{a_2b_0-b_2}{2a_2}} x^{b_0-1} \right.$$
$$\left. e^{\frac{\operatorname{artanh}\left(\frac{2a_2x+a_1}{\sqrt{a_1^2-4a_2}}\right)(a_1a_2b_0+a_1b_2-2a_2b_1)}{a_2\sqrt{a_1^2-4a_2}}} dx + c_1 \right) x^{-b_0} e^{-\frac{\operatorname{artanh}\left(\frac{2a_2x+a_1}{\sqrt{a_1^2-4a_2}}\right)(a_1a_2b_0+a_1b_2-2a_2b_1)}{a_2\sqrt{a_1^2-4a_2}}}.$$

$> simplify(subs(b_2 = \frac{2a_2b_1-a_1a_2b_0}{a_1}, \%));$

$$S(x) = (1 + a_1x + a_2x^2)^{b_0-1-\frac{b_1}{a_1}} \left(\int x^{b_0} (1 + a_1x + a_2x^2)^{-b_0+\frac{b_1}{a_1}} \left(a_1s_0 + b_0s_1 + b_1s_0 + \frac{b_0s_0}{x} + s_1\right) dx + c_1 \right) x^{-b_0}.$$

Assumption on b_2: explanation

$> \mathit{simplify}(\mathit{subs}(b_2 = \frac{2a_2b_1 - a_1a_2b_0}{a_1}, \%));$

$$S(x) = (1 + a_1x + a_2x^2)^{b_0 - 1 - \frac{b_1}{a_1}} \left(\int x^{b_0} (1 + a_1x + a_2x^2)^{-b_0 + \frac{b_1}{a_1}} \left(a_1s_0 + b_0s_1 + b_1s_0 + \frac{b_0s_0}{x} + s_1 \right) dx + c_1 \right) x^{-b_0}.$$

Theorem [André, 1989]

Let y be a power series. Then y is algebraic if and only if y is globally bounded and $\frac{dy}{dx}$ is algebraic.

Corollary

The primitive of an algebraic power series is globally bounded if and only if it is algebraic.

Integrality analysis: example

Consider a recurrence relation

$$n s_n + (2n+3) s_{n-1} + 9(n+3) s_{n-2} = 0.$$

Apply the algorithm $\implies$ $S(x) = \sum_{n \geq 0} s_n x^n$ is globally bounded if and only if $s_1 = -5 s_0$. In that case, one has

$$S(x) = s_0 (1 + 2x + 9x^2)^{-\frac{5}{2}}.$$

For which values of $s_0 \in \mathbb{Z}$ does $s_0 (1 + 2x + 9x^2)^{-\frac{5}{2}}$ lie in $\mathbb{Z}[[x]]$?

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Lemma [Pomerat, Straub, 2025]

Let $a, b \in \mathbb{Z},\ \lambda \in \mathbb{Q}$. Let k be the denominator of λ brought to the lowest terms. Then $(1+ax+bx^2)^\lambda \in \mathbb{Z}[[x]]$ if and only if

- $a, b \in k\ rad(k)\mathbb{Z}$, or
- $k = 2\kappa$ and $a, b \in \kappa\ rad(\kappa)\mathbb{Z}$ as well as $(a, b) \equiv (2, 1) \pmod 4$.

Here $rad(k)$ denotes the largest squarefree integer dividing k.
For example, $rad(4) = 2,\ rad(24) = 6$.

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In our case $a = 2, b = 9, \lambda = -\frac{5}{2}, k = 2, \kappa = 1 \implies (1+2x+9x^2)^{-\frac{5}{2}} \in \mathbb{Z}[[x]]$.
Hence, $s_0(1+2x+9x^2)^{-\frac{5}{2}} \in \mathbb{Z}[[x]]$ for all $s_0 \in \mathbb{Z}$.

Hypergeometric case

Recurrence:

$$(n + b_0)s_n + (a_1 n + b_1)s_{n-1} = 0, \quad a_1 \neq 0.$$

Closed-form expression of the generating function $S(x) = \sum_{n \geq 0} s_n x^n$:

$$x^{-b_0}(a_1 x + 1)^{-1+b_0-\frac{b_1}{a_1}} \left(s_0 b_0 \int x^{-1+b_0}(a_1 x + 1)^{\frac{-a_1 b_0 + b_1}{a_1}} dx + c_1 \right).$$

The integrand is algebraic $\implies$ the primitive is algebraic if and only it is globally bounded (*André's theorem*).

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$$S(x) = x^{-b_0}(a_1 x + 1)^{-1+b_0-\frac{b_1}{a_1}} \left(s_0 x^{b_0} {}_2F_1 \left[\begin{matrix} b_0 & b_0 - \frac{b_1}{a_1} \\ 1 + b_0 \end{matrix} ; -a_1 x \right] + c_1 \right).$$

Conditions on s_0 ensuring algebraicity of $S(x)$ can be derived from the work of Fürnsinn and Yurkevich.